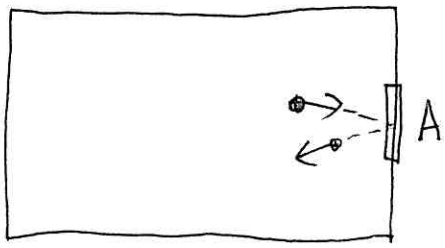
P 1.22



(a) Force due to one collision $F_1 = -m\frac{\Delta V_x}{\Delta t} = \frac{2mV_x}{\Delta t}$

$$\Delta V_x = -2V_x$$

Pressure due to N molecules colliding $P = N \cdot \frac{2mV_x}{A\Delta t}$

$$\Rightarrow N = \frac{PA\Delta t}{2mV_x} \quad (1)$$

(b) Replace v_x by $\sqrt{\overline{V_x^2}} \Rightarrow \frac{1}{2}m\overline{V_x^2} = \frac{1}{2}kT \Rightarrow \overline{V_x^2} = \frac{kT}{m}$

$$V_x \simeq \sqrt{\frac{kT}{m}}$$

(c) From (1) $\frac{\Delta N}{\Delta t} = -\frac{PA}{2mV_x}$ ((−) sign because number of molecules in container is decreasing)

$$\Rightarrow \frac{dN}{dt} = -\frac{PA}{2m}\sqrt{\frac{m}{kT}} = -\frac{A}{2}\frac{1}{\sqrt{mkT}}\frac{NkT}{V} = -\frac{A}{2V}\sqrt{\frac{kT}{m}}N$$

$$PV = NkT \Rightarrow P = \frac{NkT}{V}$$

$$\frac{dN}{dt} = -\frac{N}{\tau}, \quad \tau = \frac{2V}{A}\sqrt{\frac{m}{kT}}$$

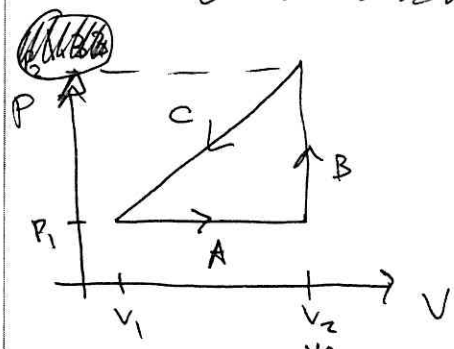
$$\Rightarrow N(t) = N(0)\,e^{-t/\tau}$$

(d) $\sqrt{\frac{kT}{m}} = \sqrt{\frac{RT}{M}} = \left(\frac{8.3\ J/mol\cdot K\ 300K}{0.03\ kg/mol}\right)^{1/2} \simeq 300\,m/s \Rightarrow$

M ↑ molar mass

$$\tau = \frac{2\cdot(0.1m)^3}{(10^{-3}m)^2\,300\,m/s} \simeq 7\,sec.$$

P 1.33



$PV = N k_B T$

(A) $W = +\int_{V_1}^{V_2} P\,dV = +P_1(V_2 - V_1) < 0$ work done on the gas

$U = N \cdot \frac{3}{2} k_B T \Rightarrow \Delta U = U_2 - U_1 = \frac{P_1}{N k_B}(V_2 - V_1) \cdot \frac{3}{2} k_B N$

$T_1 = \frac{P_1 V_1}{N k_B}, \; T_2 = \frac{P_2 V_2}{N k_B}$

$\Delta U = W + Q \Rightarrow \frac{3}{2} P_1 (V_2 - V_1) = -P_1(V_2 - V_1) + Q$

$Q_A = \frac{5}{2} P_1 (V_2 - V_1)$

(B) $\Delta V = 0 \Rightarrow W_B = 0 \quad T_i = \frac{P_1 V_2}{N k_B} \quad T_f = \frac{P_2 V_2}{N k_B}$

$\Delta U = \frac{3}{2} k_B N (T_f - T_i) = \frac{3}{2} V_2 (P_2 - P_1) > 0$

$\Rightarrow Q_B = \Delta U = \frac{3}{2} V_2 (P_2 - P_1)$

(C) $P = P_1 + \frac{V - V_1}{V_2 - V_1}(P_2 - P_1)$

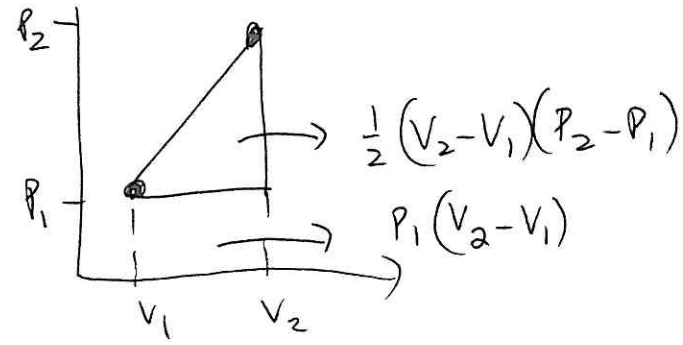
$\left.\begin{array}{ll} P_i = P_2 & V_i = V_2 \\ P_f = P_1 & V_f = V_1 \end{array}\right\} \quad W = +\int_{V_2}^{V_1} P\,dV = +\int_{V_2}^{V_1} \left(P_1 + \frac{P_2 - P_1}{V_2 - V_1}(V - V_1)\right) dV$

$= +P_1(V_1 - V_2) + \frac{P_2 - P_1}{V_2 - V_1}\left(\frac{1}{2} V^2 \Big|_{V_2}^{V_1} - V_1(V_1 - V_2)\right) =$

$= +P_1(V_1 - V_2) + \frac{P_2 - P_1}{V_2 - V_1}\left(\underbrace{\frac{1}{2} V_1^2 - \frac{1}{2} V_2^2 - V_1^2 + V_1 V_2}_{-\frac{1}{2}(V_1^2 + V_2^2 - 2V_1V_2) = -\frac{1}{2}(V_1 - V_2)^2}\right)$

$$W = +P_1(V_2 - V_1) + \frac{P_2 - P_1}{V_2 - V_1}\left(-\frac{1}{2}\right)(V_2 - V_1)^2 =$$

$$= P_1(V_2 - V_1) + \frac{1}{2}(P_2 - P_1)(V_2 - V_1) = \frac{1}{2}(P_2 + P_1)(V_2 - V_1)$$



$$U_f = \frac{3}{2} N k_B T_f = \frac{3}{2} N k_B \frac{P_1 V_1}{N k_B} = \frac{3}{2} P_1 V_1$$

$$U_i = \frac{3}{2} N k_B T_i = \frac{3}{2} N k_B \frac{P_2 V_2}{N k_B} = \frac{3}{2} P_2 V_2$$

$$\Delta U = U_f - U_i = \frac{3}{2}(P_1 V_1 - P_2 V_2)$$

$$\Rightarrow Q_C = -\frac{3}{2}(P_2 V_2 - P_1 V_1) - \frac{1}{2}(P_2 + P_1)(V_2 - V_1) =$$

$$= -\frac{3}{2}(P_2 V_2 - P_1 V_1) - \frac{1}{2}(P_2 V_2 - P_2 V_1 + P_1 V_2 - P_1 V_1)$$

$$= -2P_2 V_2 + 2P_1 V_1 + \frac{1}{2}(P_2 V_1 - P_1 V_2)$$

$$\cancel{\Sigma Q = Q_A + Q_B + Q_C = \frac{3}{2} P_1(V_2 - V_1) + \frac{3}{2} V_2(P_2 - P_1)}$$

Total work: $\Sigma W = W_A + \underset{\overset{\shortparallel}{0}}{W_B} + W_C = -P_1(V_2 - V_1) + \frac{1}{2}(P_2 + P_1)(V_2 - V_1) =$

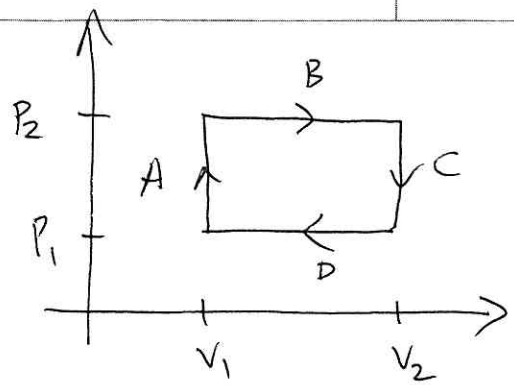
$$= \frac{1}{2}(P_2 - P_1)(V_2 - V_1)$$

Internal energy did not change as initial and final P, V are the same

$\Rightarrow \Sigma W + \Sigma Q = 0 \Rightarrow$ system does work and ~~absorbs~~ emits heat.

check $\cancel{\Sigma Q = -2P_2V_2 + 2P_1V_1 + \frac{1}{2}(P_2V_1 - P_1V_2) + \frac{3}{2}(V_2P_2 - V_2P_1) + \frac{3}{2}(P_1V_1 - P_2V_2)} =$

P 1.34



diatomic gas $\Rightarrow$ add rotational degrees of freedom

$$E = \underset{\frac{3}{2}kT}{\underset{\parallel}{K.E.}} + \underset{2\cdot\left(\frac{1}{2}kT\right)}{\underset{\parallel}{K.ROT}} = \frac{5}{2}kT.$$

$$\text{Total } \Delta W = \oint P\,dV = (P_2 - P_1)\,\overset{=0}{\Delta V_A} + P_2(V_2 - V_1) + (P_1 - P_2)\,\underset{0}{\underset{\parallel}{\Delta V_C}}$$

$$+ P_1(V_1 - V_2) = (P_2 - P_1)(V_2 - V_1).$$

Total $\Delta U = 0$ because initial and final states are the same

Total $\Delta Q = (P_2 - P_1)(V_2 - V_1)$ because $\Delta U = \Delta Q - \Delta W = 0$.

$$U = \frac{5}{2}kT\cdot N, \quad PV = NkT \Rightarrow U = \frac{5}{2}PV$$

$$\Rightarrow \left.\begin{aligned} U_B - U_A &= \frac{5}{2}(P_2 - P_1)V_1 \\ U_C - U_B &= \frac{5}{2}P_2(V_2 - V_1) \\ U_D - U_C &= \frac{5}{2}(P_1 - P_2)V_2 \\ U_A - U_D &= \frac{5}{2}P_1(V_1 - V_2) \end{aligned}\right\} \Delta U = 0$$

P1.39 $C_s = \sqrt{\frac{B}{\rho}}$ ρ = density $= m\frac{N}{V}$

$$B = \frac{\Delta P}{-\Delta V/V} = -V\frac{\Delta P}{\Delta V}$$

(a) $PV = NkT \Rightarrow$ for $T = $ const (Isothermal)

$$dP \cdot V + PdV = 0 \Rightarrow \frac{dP}{dV} = -\frac{P}{V} = -\frac{NkT}{V^2}$$

$$\Rightarrow B_{Iso} = -V\,\frac{-NkT}{V^2} = \frac{NkT}{V} = P$$

Adiabatic $PV^\gamma = \text{const} \Rightarrow dP \cdot V^\gamma + P\gamma V^{\gamma-1}dV = 0$

$$\frac{dP}{dV} = -\gamma\frac{PV^{\gamma-1}}{V^\gamma} = -\gamma\frac{P}{V} = -\gamma\frac{NkT}{V^2}$$

$$\Rightarrow B_{ad} = -V\left(-\gamma\frac{NkT}{V^2}\right) = \gamma\frac{NkT}{V} = \gamma P.$$

$B_{adiab} > B_{Isoth}$

$\gamma P > P$

(b) Thermal conductivity of air is low $\Rightarrow$ pressure wave moves much quicker than heat $\Rightarrow$ adiabatic process is more appropriate

(c) $$\Rightarrow C_s = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma P}{mP/kT}} = \sqrt{\frac{\gamma kT}{m}} = \sqrt{\gamma\frac{RT}{M}} \approx 350\ \text{m/s}$$

for $\gamma = 1.4$

$$\rho = m\frac{N}{V} = m\frac{P}{kT}$$

(d) Speed of sound depends on temperature, not pressure, as part (c) shows. So the T variation is probably not too large to de-tune the bag pipe.